

**CODEALPHA ROBOTICS & AUTOMATION
INTERNSHIP**

REPORT FOR

Automation System Design

Of

Automated Packaging System

Submitted To

Code technologies (Code Alpha)

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AUTOMATION SYSTEM:

Title

Design of an Automated Packaging System Using Sensors and Conveyor Belt

Objective

To design an automated packaging system capable of detecting products, transporting them through a conveyor belt, packaging them automatically, and delivering the finished package to the output station.

Introduction

Automation plays a significant role in modern industries by reducing human intervention, increasing productivity, and improving product quality. Automated packaging systems are widely used in manufacturing industries, food processing plants, pharmaceutical industries, e-commerce warehouses, and logistics centers.

This project presents the design of an Automated Packaging System that uses sensors, conveyor belts, and control systems to perform packaging operations automatically.

Components Used

1. Conveyor Belt
2. DC Motor
3. Power Supply
4. Proximity Sensor (IR Sensor)
5. PLC / Microcontroller
6. Packaging Station
7. Sealing Unit
8. Product Box
9. Control Panel

Working Principle

The conveyor belt continuously transports products from the loading station toward the packaging area.

A proximity sensor is installed near the packaging station to detect incoming products.

When a product reaches the sensor:

1. The sensor sends a signal to the PLC.
2. The conveyor motor stops.
3. The product is positioned inside the packaging area.
4. The packaging unit places the product into a box.
5. The sealing mechanism closes and seals the package.

6. The conveyor restarts.
7. The packaged product is moved to the output station.

This process repeats automatically for every product.

Flowchart:

Automated Packaging System

End-to-end flowchart: product intake → inspection → packaging → labeling → dispatch



